FLATNESS OF A MODULE OVER A GLUEING PAIR IS DETECTED BY ITS TWO LOCALIZATIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

Editorial note. The author's sentence 'the action on f on Tor_i^R(...)' in the last line is reproduced unchanged. The seven references inside the counted proof span are recorded as non-load-bearing navigation pointers: algebra-lemma-flat-base-change, algebra-lemma-flat-tor-zero, algebra-lemma-characterize-flat, algebra-theorem-lazard and example-tor-change-rings resolve to the project's cross-chapter tag list, while remark-what-you-get-for-general-modules and lemma-BL-faithful are same-chapter labels resolved to their \label lines in the pinned source (25676 and 25149).

Allowed standard prerequisites (editorial).

- Glueing pairs (R -> R', f), glueable modules and the glued module M-tilde. fixed earlier in the same section, used at proof lines 25709-25710 and 25720. Supplies the reduction from an arbitrary M to a glueable one, which is the first half of the converse direction.
- The flatness criterion via Tor_1^R(M, N) = 0, and Lazard's theorem that a flat module is a filtered colimit of finite free modules. used at proof lines 25729-25730 and 25751-25752. Converts the problem into a Tor-vanishing statement and lets the f-action be transferred to the base change.
- The change-of-rings spectral sequence for Tor and the exact sequence 0 -> R -> R' oplus R_f -> (R')_f -> 0. used at proof lines 25733-25750. Computes Tor_i^R(M tensor_R R', N) from Tor_i^R(R', N) and produces the f-action statement.

Source reference. Original lemma 00M5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 811–811. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 00HI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 644–644. Cross-chapter navigation pointer into the

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Source reference. Original lemma 00HL: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 647–647. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 058G: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6779–6779. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original remark 15192: sources/stacks-more-algebra/pinned-original-source.tex, source lines 15192–15192. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 25149: sources/stacks-more-algebra/pinned-original-source.tex, source lines 25149–25149. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original remark 25676: sources/stacks-more-algebra/pinned-original-source.tex, source lines 25676–25676. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let $(R \to R', f)$ be a glueing pair. Let M be an R-module which is not necessarily glueable for $(R \to R', f)$. Then M is flat over R if and only if $M \otimes_R R'$ is flat over R' and M_f is flat over R_f.*

Proof. One direction of the lemma follows from Algebra, Lemma 00HI. For the other direction, assume $M \otimes_R R'$ is flat over R' and M_f is flat over R_f. Let $\tilde{M}$ be as in Remark 25676. If $\tilde{M}$ is flat over R, then applying Algebra, Lemma 00HL to the short exact sequence $0 \to \mathrm{Ker}(M \to \tilde{M}) \to M \to \tilde{M} \to 0$ we find that $\mathrm{Ker}(M \to \tilde{M}) \otimes_R (R' \oplus R_f)$ is zero. Hence $M = \tilde{M}$ by Lemma 25149 and we conclude. In other words, we may replace M by $\tilde{M}$ and assume M is glueable for $(R \to R', f)$. Let N be a second R-module. It suffices to prove that $\mathrm{Tor}_1^R(M, N) = 0$, see Algebra, Lemma 00M5.

The long exact sequence of Tors associated to the short exact sequence $0 \to R \to R' \oplus R_f \to (R')_f \to 0$ and N gives an exact sequence

$$0 \to \mathrm{Tor}_1^R(R', N) \to \mathrm{Tor}_1^R((R')_f, N)$$

and isomorphisms $\mathrm{Tor}_i^R(R', N) = \mathrm{Tor}_i^R((R')_f, N)$ for $i \geq 2$. Since $\mathrm{Tor}_i^R((R')_f, N) = \mathrm{Tor}_i^R(R', N)_f$ we conclude that f is a nonzerodivisor on $\mathrm{Tor}_1^R(R', N)$ and invertible on $\mathrm{Tor}_i^R(R', N)$ for $i \geq 2$. Since $M \otimes_R R'$ is flat over R' we have

$$\mathrm{Tor}_i^R(M \otimes_R R', N) = (M \otimes_R R') \otimes_{R'} \mathrm{Tor}_i^R(R', N)$$

by the spectral sequence of Example 15192. Writing $M \otimes_R R'$ as a filtered colimit of finite free R'-modules (Algebra, Theorem 058G) we conclude that f is a nonzerodivisor on $\mathrm{Tor}_1^R(M \otimes_R R', N)$ and invertible on $\mathrm{Tor}_i^R(M \otimes_R R', N)$. Next, we consider the exact sequence $0 \to M \to M \otimes_R R' \oplus M_f \to M \otimes_R (R')_f \to 0$ coming

from the fact that M is glueable and the associated long exact sequence of Tor. The relevant part is

$$\begin{array}{ccccc} \mathrm{Tor}_1^R(M,N) & \longrightarrow & \mathrm{Tor}_1^R(M\otimes_R R',N) & \longrightarrow & \mathrm{Tor}_1^R(M\otimes_R (R')_f,N) \\ & \searrow & & & \\ & & \mathrm{Tor}_2^R(M\otimes_R R',N) & \longrightarrow & \mathrm{Tor}_2^R(M\otimes_R (R')_f,N) \end{array}$$

We conclude that $\mathrm{Tor}_1^R(M,N)=0$ by our remarks above on the action on f on $\mathrm{Tor}_i^R(M\otimes_R R',N)$. □